phonokit

A toolkit to create phonological representations

VERSION 0.5.1 • MARCH 2026

Guilherme D. Garcia | gdgarcia.ca

UNIVERSITÉ LAVAL

A new language for typesetting documents

Typst is a new language to typeset documents. It's modern, light, fast and intuitive. Visit typst.app to use their online editor (also check out their excellent tutorials). These slides were made using Typst (a custom version of the Diatypst template).

A new package for phonology

- A Typst package for phonology (Garcia, 2026)
- **Idea**: generate phonological representations (IPA, prosody, SPE, OT, etc.)
- Current version: `0.5.1`

phonokit

What does LaTeX offer to phonologists?

Designed for phonology:

- IPA with `tipa`
- Tableaux with `ot-tableau`
- SPE rules and matrices with math mode and specialized packages, such as `phonrule`
- Vowel trapezoids with `vowel`

Adapted to phonology:

- Numbered examples with `linguex` (among others)
- Trees with `tikz` and `tikz-qtree` (meant for syntax)
- Non-linear representations in general with `tikz`

What can be improved?

- **Goal:** offer specialized functions that **automate** phonological representations with **minimal syntax**

IPA	OT	MaxEnt	(N)HG	autosegmental
sonority	examples	ToBI	SPE	consonants
vowels	grids	syllables	feet	PWds
moras	multi-tier	Hasse	simulations	calculations

Table 1: Components currently covered by **phonokit**

☞ **phonokit** improves what is already there in LaTeX and adds more

Phonetic transcription

☞ Charis is the default font, but you can alter it (these slides use Libertinus Sans)

UTF-8 is supported natively in Typst 😌, so you could add phonetic symbols directly (easy but slow)

```
#ipa("['pE.tSi]")
```

[ˈpɛ.tʃi]

```
#ipa("['b2R \\schwar ,flaI]")
```

[ˈbʌɾɚˌflaɪ]

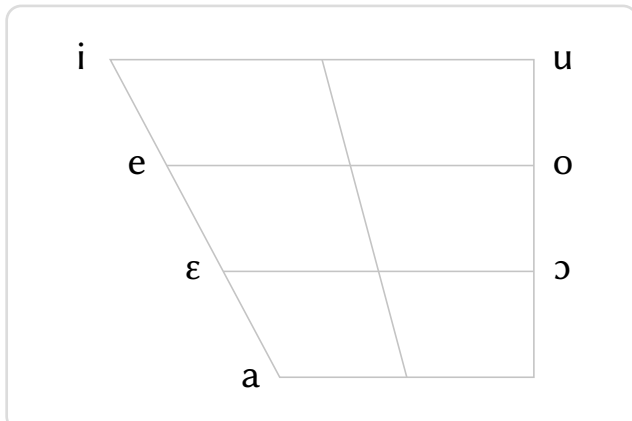
```
#ipa("[ma.' \\nh a.na]")
```

[ma.ˈɲa.na]

Phonemic inventories: vowels & consonants

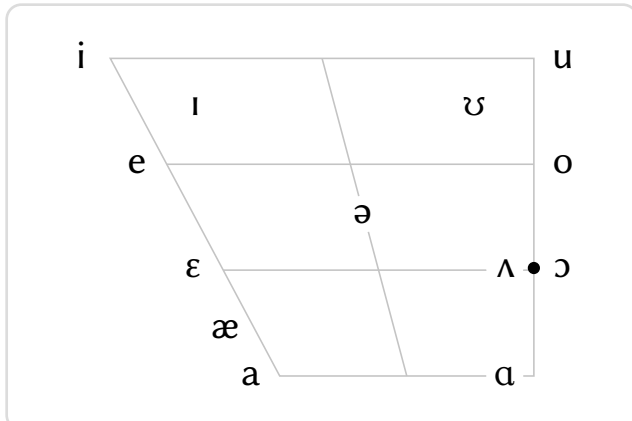
- Pre-defined languages or custom inventories

```
#vowels("portuguese")
```



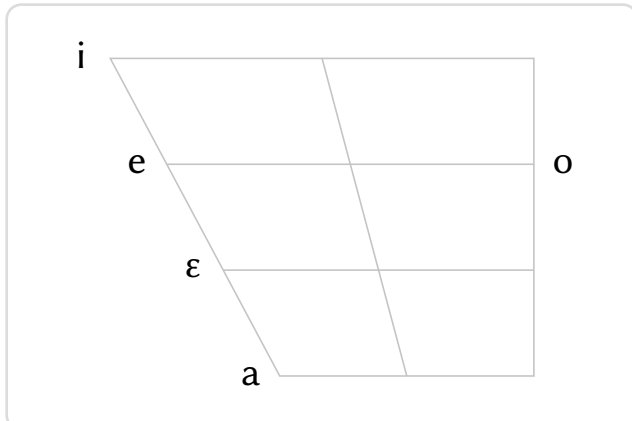
Phonemic inventories: vowels & consonants (ii)

```
#vowels("english")
```



Phonemic inventories: vowels & consonants (iii)

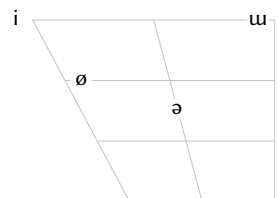
```
#vowels("aeioE")
```



Phonemic inventories: vowels & consonants (iv)

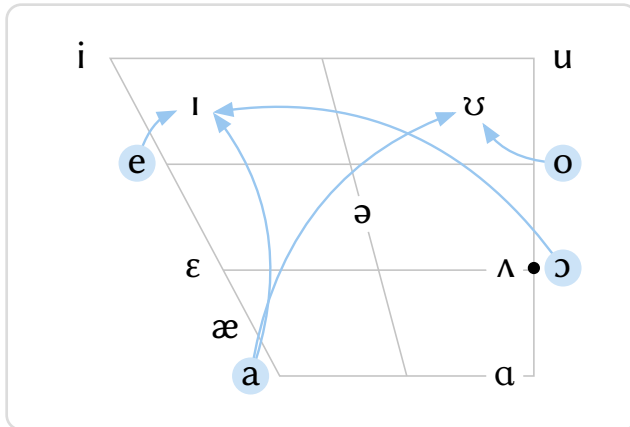
- Outputs are easy to scale up/down

```
#vowels("\\o iW@", scale: 0.4)
```



Phonemic inventories: vowels & consonants (v)

```
#vowels(
    "english",
    arrows: (
      ("a", "U"),
      ("a", "I"),
      ("e", "I"),
      ("O", "I"),
      ("o", "U"),
    ),
    arrow-color: blue.lighten(60%),
    curved: true,
    highlight: ("a", "e", "o", "O"),
    highlight-color: blue.lighten(80%),
  )
```



Phonemic inventories: vowels & consonants (vi)

```
#consonants("portuguese", scale: 0.65)
```

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d				k g			
Nasal	m			n			ɲ				
Trill											
Tap or Flap				ɾ							
Fricative		f v		s z	ʃ ʒ			x			
Lateral fricative											
Approximant	w						j	w			
Lateral approximant				l			ʎ				

Phonemic inventories: vowels & consonants (vii)

```
#consonants("italian", scale: 0.55)
```

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d				k g			
Nasal	m			n			ɲ				
Trill				r							
Tap or Flap											
Fricative		f v		s z	ʃ						
Lateral fricative											
Approximant							j				
Lateral approximant				l			ʎ				

Phonemic inventories: vowels & consonants (viii)

```
#consonants("bpmnNKslL", scale: 0.45)
```

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b										
Nasal	m			n				ŋ			
Trill											
Tap or Flap											
Fricative				s					ʁ		
Lateral fricative											
Approximant											
Lateral approximant				l			ʎ				

SPE

- Feature matrices for a given phoneme; matrices for rules + helper functions such as `#blank()`

```
#feat-matrix("\\ae")
```

/æ/

$$\begin{bmatrix} +\text{syllabic} \\ -\text{consonantal} \\ +\text{sonorant} \\ +\text{continuant} \\ +\text{voice} \\ -\text{high} \\ +\text{low} \\ +\text{front} \\ -\text{back} \\ -\text{round} \end{bmatrix}$$

```
#feat("+son", "-approx") #a-r #feat(alpha +
[#smallcaps("place")]) / #blank()\]#sub[#sigma] #feat("-
son", "-cont", "-del rel", alpha + [#smallcaps("place")])
```

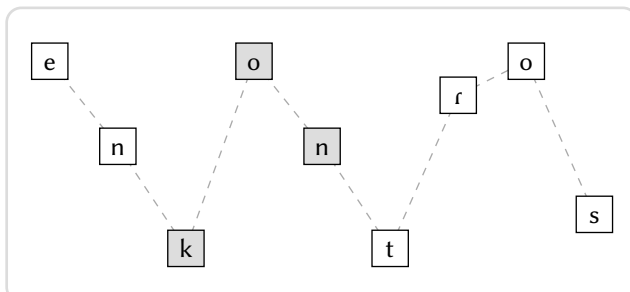
$$\begin{bmatrix} +\text{son} \\ -\text{approx} \end{bmatrix} \rightarrow \left[\alpha\text{PLACE}\right] / \underline{\qquad}\,]_{\sigma} \begin{bmatrix} -\text{son} \\ -\text{cont} \\ -\text{del rel} \\ \alpha\text{PLACE} \end{bmatrix}$$

Visualizing sonority profiles

- Visual representation of the sonority principle

(Parker, 2011)

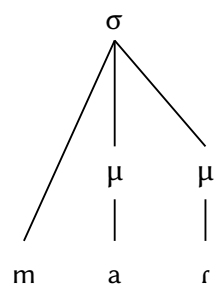
```
#sonority("en.kon.tRos", scale: 0.6)
```



Prosodic representation

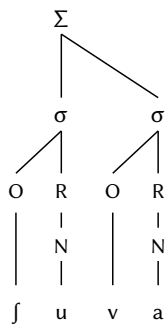
```
#syllable("maR", scale: 0.9)
```

```
#mora("maR", coda: true)
```

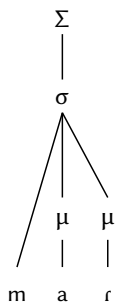


Prosodic representation

```
#foot("'Su.va", scale: 0.8)
```

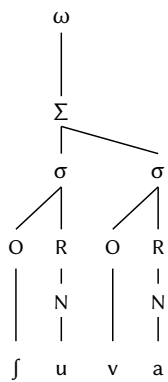



```
#foot-mora("maR", coda: true, scale: 0.8)
```

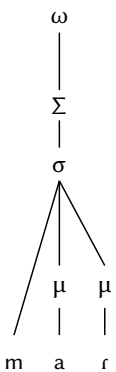


Prosodic representation

```
#word("('Su.va)", scale: 0.8)
```

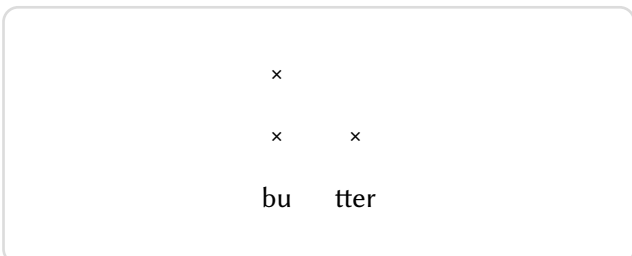



```
#word-mora("(maR)", coda: true, scale: 0.8)
```

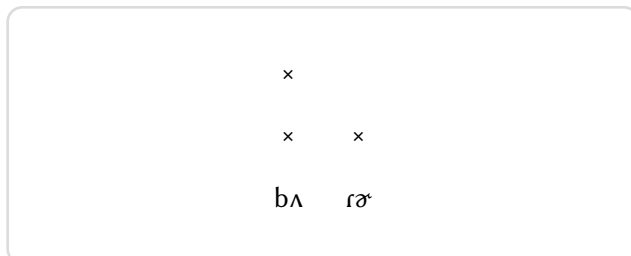


Prosodic representation: metrical grids

```
#met-grid("bu2.tter1")
```

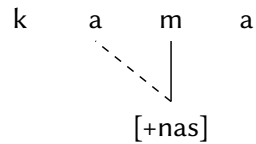



```
#met-grid(("b2", 2), ("R \\schwar", 1))
```



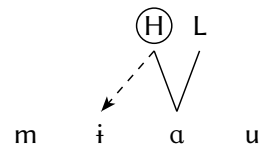
Autosegmental phonology

```
#autoseg(
    ("k", "a", "m", "a"),
    features: ("", "", "[+nas]", ""),
    links: ((2,1),),
    spacing: 1.0,
    arrow: false
    )
```



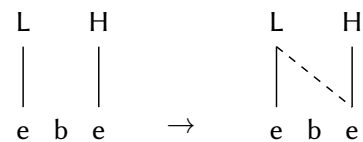
Autosegmental phonology

```
#autoseg(
      ("m", "1", "A", "u"),
      features: ("", "", ("H", "L"), ""),
      tone: true,
      links: (((2, 0), 1),),
      highlight: ((2, 0),),
      spacing: 1.0,
      arrow: true,
    )
```



Autosegmental phonology

```
#autoseg(
      ("e", "b", "e"),
      features: ("L", "", "H"),
      spacing: 0.5,
      tone: true,
      gloss: [])
   #a-r
   #autoseg(
      ("e", "b", "e"),
      features: ("L", "", "H"),
      links: ((0, 2),),
      spacing: 0.5,
      // arrow: true,
      tone: true,
      gloss: [èbě _pumpkin_])
```



èbě *pumpkin*

Prosody with ToBi

- Examples from Zsiga (2024)

```
You're a we#int("*L")rewolf?#h(1em)#int("H%",
line: false)
```

*L H%

You're a werewolf?

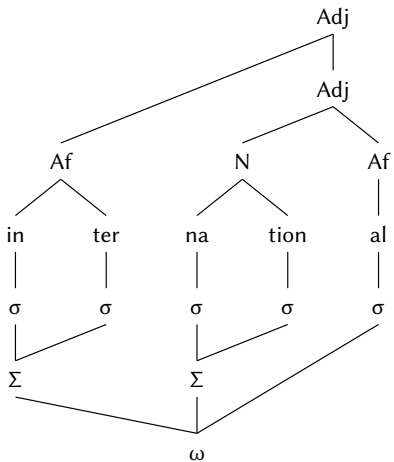
```
I'm a wer#int("*H")ewolf.#h(1em)#int("L%",
line: false)
```

*H L%

I'm a werewolf.

☞ The function `#int()` works well with numbered examples — see manual and slide 36 ↓

Multi-tier representations

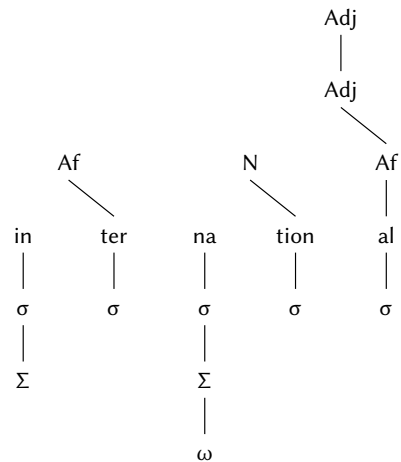


- Function `#multi-tier()` : **very flexible**
- Wide range of arguments based on a grid architecture
- Helper: temporary grid with coordinates
- Figure adapted from Booij (2012)

☞ Let's unpack this figure and its code

Multi-tier representations

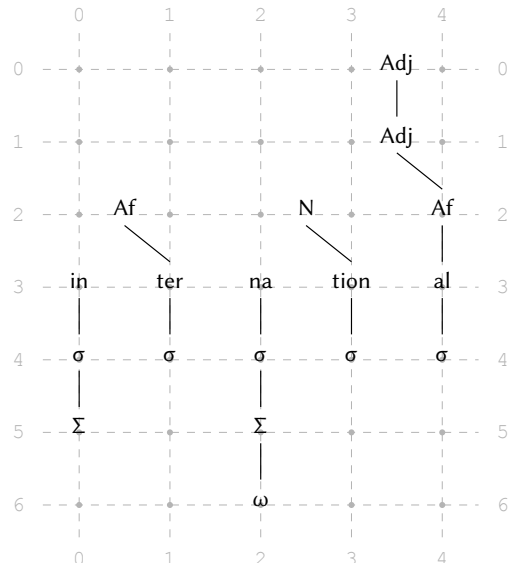
```
#multi-tier(
  show-grid: false,
  levels: (
    ("", "", "", "", ("Adj", 3.5)),
    ("", "", "", "", ("Adj", 3.5)),
    ("", ("Af", 0.5), "", ("N", 2.5), "Af"),
    ("in", "ter", "na", "tion", "al"),
    ("sigma", "sigma", "sigma", "sigma", "sigma"),
    ("Sigma", "", "Sigma", "", ""),
    ("", "", "omega", "", ""),
  ),
  scale: 0.8,
 )
```



- Any element projects **one** line/link by default (this can be deleted later with `delinks`)

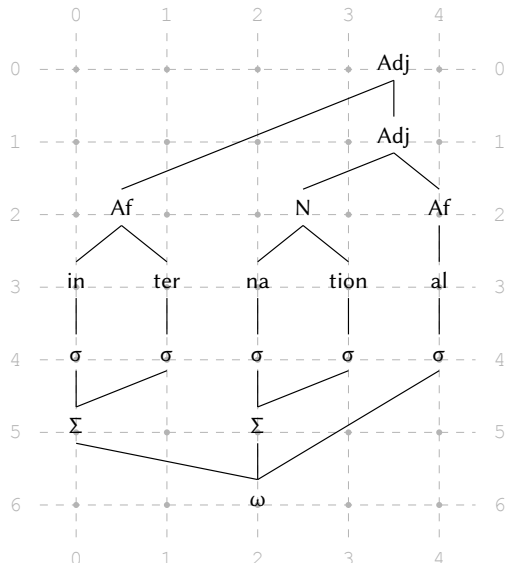
Multi-tier representations

```
#multi-tier(
      show-grid: true, // ← HELPER GRID
      levels: (
        ("", "", "", "", ("Adj", 3.5)),
        ("", "", "", "", ("Adj", 3.5)),
        ("", ("Af", 0.5), "", ("N", 2.5), "Af"),
        ("in", "ter", "na", "tion", "al"),
        ("sigma", "sigma", "sigma", "sigma", "sigma"),
        ("Sigma", "", "Sigma", "", ""),
        ("", "", "omega", "", ""),
      ),
      scale: 0.8,
    )
```



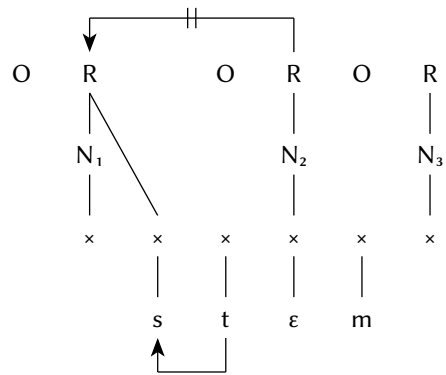
Multi-tier representations

```
#multi-tier(
      show-grid: true,
      levels: (
        ("", "", "", "", ("Adj", 3.5)),
        ("", "", "", "", ("Adj", 3.5)),
        ("", ("Af", 0.5), "", ("N", 2.5), "Af"),
        ("in", "ter", "na", "tion", "al"),
        ("sigma", "sigma", "sigma", "sigma", "sigma"),
        ("Sigma", "", "Sigma", "", ""),
        ("", "", "omega", "", ""),
      ),
      scale: 0.8,
      links: (
      ((0, 4), (2, 1)), // Adj → Af
      ((1, 4), (2, 3)), // Adj → N
      ((2, 1), (3, 0)), // Af → in
      ((2, 3), (3, 2)), // N → na
      ((5, 0), (4, 1)), // Ft → Syl
      ((5, 2), (4, 3)), // Ft → Syl
      ((6, 2), (5, 0)), // PWd → Ft
      ((6, 2), (4, 4)), // PWd → Ft
      ),
    )
```



Government Phonology

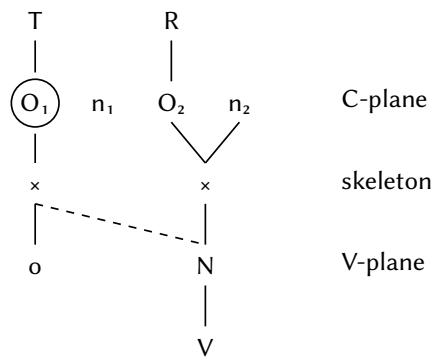
```
#multi-tier(
      levels: (
        ("O", "R", "", "O", "R", "O", "R"),
        ("", "N1", "", "", "N2", "", "N3"),
        ("", "x", "x", "x", "x", "x", "x"),
        ("", "", "s", "t", "E", "m", ""),
      ),
      links: (
        ((0, 1), (2, 2)),
      ),
      ipa: (3,),
      arrows: (
        ((3, 3), (3, 2)),
        ((0, 4), (0, 1)),
      ),
      arrow-delinks: (
        (1,)
      ),
      spacing: 1, scale: 0.8,
    )
```



Adapted from Goad (2012)

CV phonology

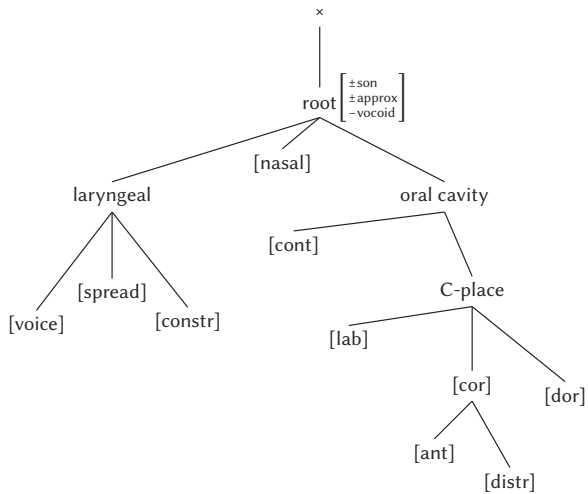
```
#multi-tier(
 levels: (
   ("T", "", "R", ""),
   ("O1", "n1", "O2", "n2"),
   ("x", "", ("x", 2.5), ""),
   ("o", "", ("N", 2.5), ""),
   ("", "", ("V", 2.5), ""),
 ),
 links: (((1, 3), (2, 2)),),
 dashed: (((2, 0), (3, 2)),),
 level-spacing: 1.2,
 highlight: ((1, 0),),
 spacing: 1,
 stroke-width: 0.7pt,
 tier-labels: (
   (1, "C-plane"),
   (2, "skeleton"),
   (3, "V-plane"),),
 scale: 1,
)
```



Adapted from Carvalho (2017)

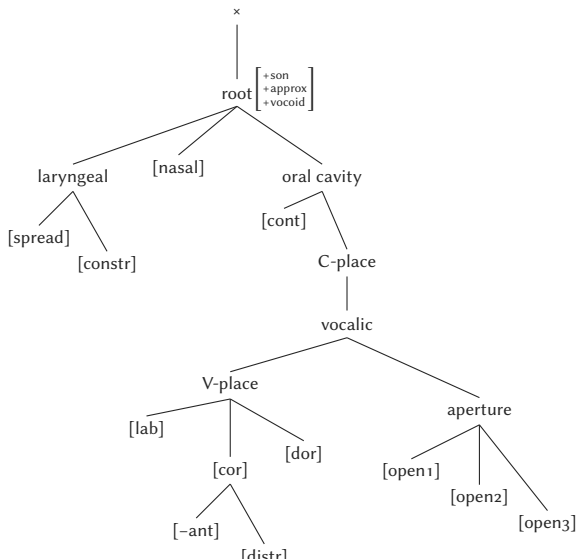
Feature Geometry: consonants (full structure)

```
#geom(
  root: ("±son", "±approx", "-vocoid"),
  spread: true,
  constricted: true,
  nasal: true,
  voice: true,
  labial: true,
  coronal: true, // ← implied
  anterior: true,
  distributed: true,
  dorsal: true,
  continuant: true,
)
```



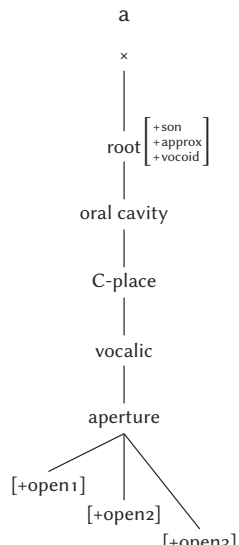
Feature Geometry: vocoids (full structure)

```
#geom(
  root: ("+son", "+approx", "+vocoid"),
  spread: true,
  constricted: true,
  nasal: true,
  vplace: true, // ← implied
  labial: true,
  aperture: (true, true, true),
  coronal: true,
  anterior: "-",
  distributed: true,
  dorsal: true,
  continuant: true,
  scale: 0.9,
)
```



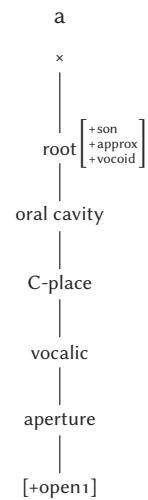
Feature Geometry: presets

```
#geom(ph: "a")
```



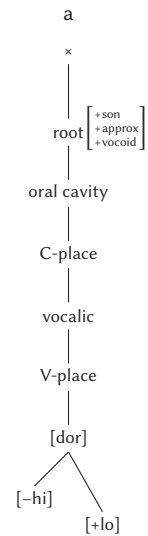
Feature Geometry: presets (ii)

```
#geom(ph: "a",
  aperture: ("+",))
```



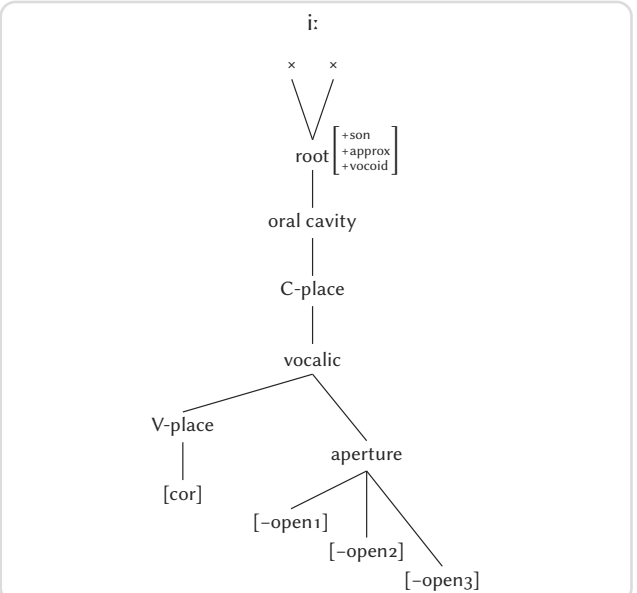
Feature Geometry: presets (iii)

```
#geom(ph: "a",
  model: "sagey",
  scale: 0.9)
```



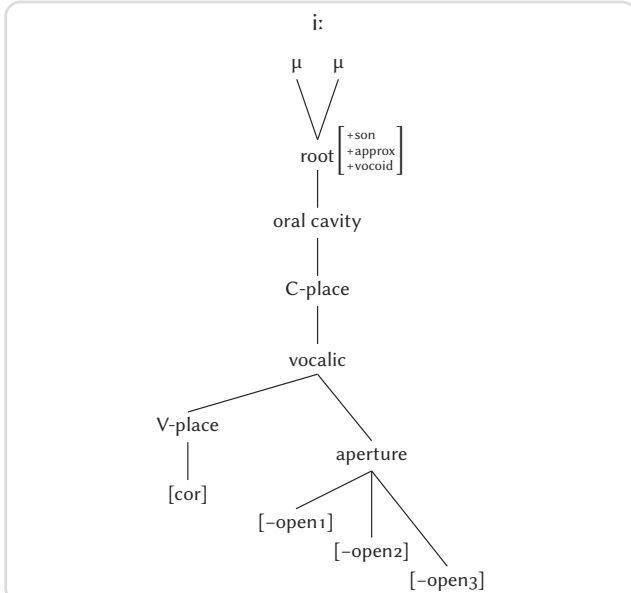
Feature Geometry: presets (iv)

```
#geom(ph: "i:")
```



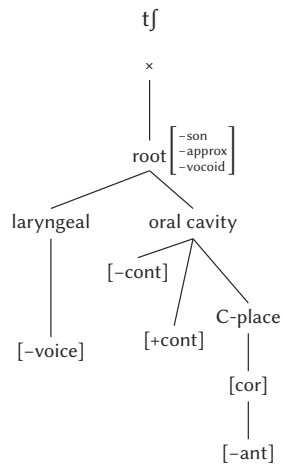
Feature Geometry: presets (v)

```
#geom(ph: "i:",
  timing: ("mora", "mora"))
```



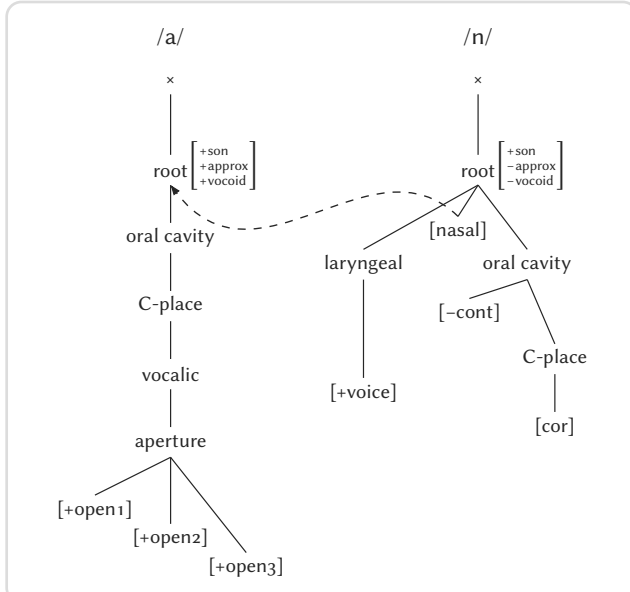
Feature Geometry: presets (vi)

```
#geom(ph: "tS")
```



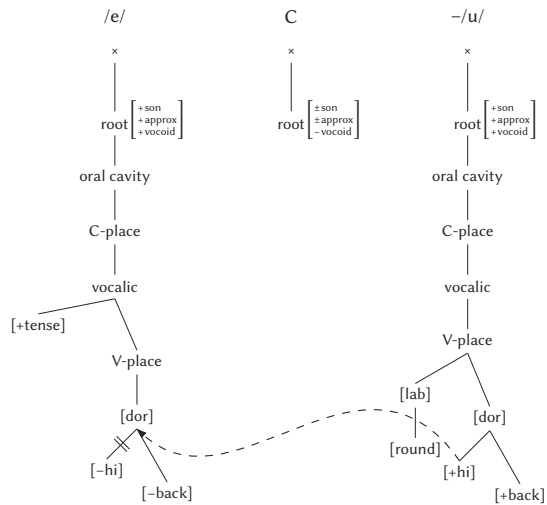
Multiple trees: processes

```
#geom-group(
    (ph: "/a/"),
    (ph: "/n/"),
    arrows: (
      (from: "nasal2",
       to: "root1",
       ctrl: (1.1, -1.5)),
       ),
    curved: true, // ← implied
    gap: 1,
  )
```



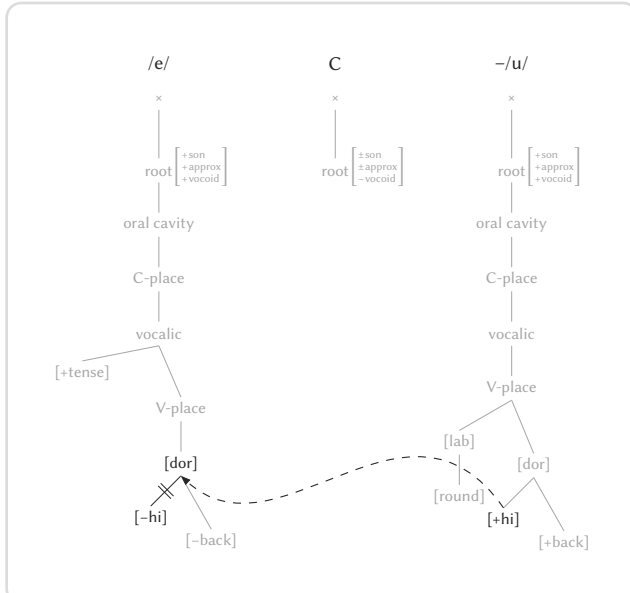
Multiple trees: processes (ii)

```
#geom-group(
  (ph: "/e/"),
  (ph: "\\C"),
  (ph: "/u/", prefix: "-"),
  arrows: (
    (from: "high3",
     to: "dorsal1",
     ctrl: (2.3, -1.5)),
  ),
  delinks: ("high1",),
  gap: 1.5,
  model: "sagey",
  scale: 0.8
)
```



Multiple trees: processes (iii)

```
#geom-group(
  (ph: "/e/"),
  (ph: "\\C"),
  (ph: "/u/", prefix: "-"),
  arrows: (
    (from: "high3",
     to: "dorsal1",
     ctrl: (2.3, -1.5)),
  ),
  delinks: ("high1",),
  highlight: (
     "high1", "dorsal1", "high3"
  ),
  gap: 1.5,
  model: "sagey",
  scale: 0.8
)
```



OT

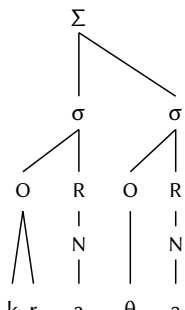
- Dynamic tableaux with auto shading (optional)

```
#tableau(
    input: "kraTa",
    candidates: ("kra.Ta", "ka.Ta", "ka.ra.Ta"),
    constraints: ("Max", "Dep", "*Complex"),
    violations: (
    ("", "", "*"),
    ("*!", "", ""),
    ("", "*!", ""), ),
    winner: 0,
    dashed-lines: (0,),
    shade: true // ← auto shading after !
)
```

/kraθa/	Max	Dep	*Complex
☞ a. kra.θa			*
b. ka.θa	*!		
c. ka.ra.θa		*!	

OT

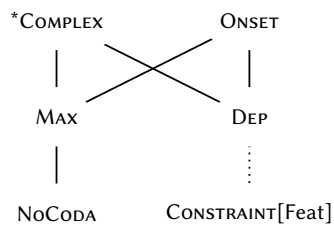
- Prosodic representation of candidates (subset of previous tableau shown here) — see manual
- Existing functions can be combined: `#foot()` inside `#tableau()`

/kraθa/	MAX	DEP	*COMPLEX
☞ 			*

Hasse diagrams

- Visualizing OT rankings with minimal syntax and with automatic small caps

```
#hasse(
    (
    ("*Complex", "Max", 0),
    ("*Complex", "Dep", 0),
    ("Onset", "Max", 0),
    ("Onset", "Dep", 0),
    ("Max", "NoCoda", 1),
    ("Dep", "Constraint[Feat]", 1, "dotted"),
    ),
    node-spacing: 3,
)
```



Harmonic Grammar

- Weights and violation counts are used to automatically compute harmony scores (h_i)

```
#hg(
    input: "kraTa",
    candidates: ("[kra.Ta]", "[ka.Ta]", "[ka.ra.Ta]"),
    constraints: ("Max", "Dep", "*Complex"),
    weights: (2.5, 1.8, 0.5),
    violations: (
      (0, 0, -1),
      (-1, 0, 0),
      (0, -1, 0),
    ),
    scale: 0.8,
    )
```

	$w = 2.5$	$w = 1.8$	$w = 0.5$	
/kraθa/	MAX	DEP	*COMPLEX	h_i
[kra.θa]	0	0	−1	−0.5
[ka.θa]	−1	0	0	−2.5
[ka.ra.θa]	0	−1	0	−1.8

Noisy Harmonic Grammar

- Probabilities simulated (Monte Carlo) based on `num-simulations` (default: `1000`)
- $\varepsilon_i \rightarrow$ single noise sample shown for illustration — not used by P_i

```
#nhg(
  input: "kraTa",
  candidates: ("[kra.Ta]", "[ka.Ta]",
"[ka.ra.Ta]"),
  constraints: ("Max", "Dep", "*Complex"),
  weights: (2.5, 1.8, 0.5),
  violations: (
    (0, 0, -1),
    (-1, 0, 0),
    (0, -1, 0),
  ),
  scale: 0.7,
  letters: true,
)
```

		$w = 2.5$	$w = 1.8$	$w = 0.5$			
	/kraθa/	Max	Dep	*Complex	h_i	ε_i	P_i
a.	[kra.θa]	0	0	−1	−0.5	−0.47	0.778
b.	[ka.θa]	−1	0	0	−2.5	−0.45	0.05
c.	[ka.ra.θa]	0	−1	0	−1.8	−1.35	0.172

MaxEnt

- MaxEnt tableaux with automatic calculation and optional probability visualization

	$w = 2.5$	$w = 1.8$	$w = 0.5$			
/kraθa/	MAX	DEP	*COMPLEX	h_i	e^{-h_i}	P_i
[kra.θa]	0	0	1	0.5	0.607	0.71
[ka.θa]	1	0	0	2.5	0.082	0.096
[ka.ra.θa]	0	1	0	1.8	0.165	0.194

	$w = 2.5$	$w = 1.8$	$w = 0.5$			
/kraθa/	MAX	DEP	*COMPLEX	h_i	e^{-h_i}	P_i
[kra.θa]	0	0	1	0.5	0.607	0.71
[ka.θa]	1	0	0	2.5	0.082	0.096
[ka.ra.θa]	0	1	0	1.8	0.165	0.194

MaxEnt

- MaxEnt tableaux with automatic calculation and optional probability visualization

```
#maxent(
    input: "kraTa",
    candidates: ("[kra.Ta]", "[ka.Ta]", "[ka.ra.Ta]"),
    constraints: ("Max", "Dep", "*Complex"),
    weights: (2.5, 1.8, 0.5),
    violations: (
      (0, 0, 1),
      (1, 0, 0),
      (0, 1, 0),
    ),
    visualize: true, // ← visualization
    sort: true, // ← sort candidates by probability
)
```

MaxEnt

- You can also easily sort candidates by P_i with `sort: true` as of version `0.4.1`

	$w = 2.5$	$w = 1.8$	$w = 0.5$			
/kraθa/	MAX	DEP	*COMPLEX	h_i	e^{-h_i}	P_i
[kra.θa]	0	0	1	0.5	0.607	0.71
[ka.θa]	1	0	0	2.5	0.082	0.096
[ka.ra.θa]	0	1	0	1.8	0.165	0.194

	$w = 2.5$	$w = 1.8$	$w = 0.5$			
/kraθa/	MAX	DEP	*COMPLEX	h_i	e^{-h_i}	P_i
[kra.θa]	0	0	1	0.5	0.607	0.71
[ka.ra.θa]	0	1	0	1.8	0.165	0.194
[ka.θa]	1	0	0	2.5	0.082	0.096

Numbered examples

- Phonology-friendly numbered examples: (1a) and (1b) are easy to reference
- Alignment is guaranteed given **table** structure; optional caption for table of contents

```
#show: ex-rules // ← this must be added to your doc
#ex(caption: "A phonology example")[
  #table(
    columns: 5, // ← where we may specify widths
    stroke: none,
    align: left,
    [#ex-num-label()], [#subex-label()<ex-anba>], [#ipa("/anba/")], [#a-r], [#ipa("[amba]")],
    [], [#subex-label()<ex-anka>], [#ipa("/anka/")], [#a-r], [#ipa("[aNka]")],
  )]
```

(1) a. /anba/ → [amba]

b. /anka/ → [aŋka]

Numbered examples

- Here's an example of a **numbered example with ToBI**
- You can refer to (2a) and (2b). As of version `0.4.6`, `title` is also available for examples:

(2) Autosegmental transcription of intonation in English (Zsiga, 2024)

a. *L H%
You're a werewolf?

b. *H L%
I'm a werewolf.

go back to slide 16 ↑

FAQ

Common questions

1. Do I need to adopt Typst to take advantage of **phono*kit***?
2. Can I completely replace LaTeX with Typst in 2026?
3. How about my `bib` references?
4. What *can't* I do with Typst yet?
5. What editor/IDE do I need to use it?

FAQ & final thoughts

1. No. You can export outputs as `PNG` and use them in LaTeX, Word, etc. Pair it with `oxipng` for tiny file sizes. See workflow example in Garcia (2026, appendix).
2. That depends. Journals will take a while to accept `typ`, and very few people know Typst. But you don't have to choose: they're two useful tools/languages. If you work in phonology, you *could* probably use Typst 99% of the time. In syntax, LaTeX still offers more when it comes to trees.
3. They work with Typst. So your workflow will not be affected.
4. LaTeX is much older, so it has **many** more packages. What you can/can't do depends on what packages your workflow requires.
5. VS Code, Positron, NeoVim, etc. Use `tinymist` as your extension/plugin.

References

Booij, G. (2012). *The grammar of words: An introduction to linguistic morphology* (3rd ed.). Oxford University Press.

Carvalho, J. B. d. (2017). Deriving sonority from the structure, not the other way round: A Strict CV approach to consonant clusters. *The Linguistic Review*, *34*(4), 589–614.

Garcia, G. D. (2026,). *phonokit: a toolkit to create phonological representations in Typst*. Zenodo. https://doi.org/10.5281/zenodo.18434478

Goad, H. (2012). sC clusters are (almost always) coda-initial. *Linguistic Review*, *29*(3).

Parker, S. (2011). Sonority. In M. van Oostendorp, C. J. Ewen, E. Hume, & K. Rice (Eds.), *The Blackwell Companion to Phonology: The Blackwell Companion to Phonology* (pp. 1160–1184). Wiley Online Library. https://doi.org/10.1002/9781444335262.wbctp0049

Zsiga, E. C. (2024). *The sounds of language: An introduction to phonetics and phonology* (2nd ed.). John Wiley & Sons.

Appendix

From LaTeX to Typst

☞ If you're coming from LaTeX, here are some equivalent packages:

LaTeX	Typst
tipa	phonokit
tikz	cetz
tikz-qtree	syntree
linguex	eggs
glossaries	glossarium